

1. Bijections

1) $1, 10, 19, \dots, 2026 \mid +8$

$9, 18, 27, \dots, 2034 \mid :9$

$1, 2, 3, \dots, 226 \Rightarrow 226 \text{ numbers}$

$100: 1 + 99 \cdot 9 = 891 + 1 = 892$

2) $1^2, 2^2, \dots, 45^2$ $1, 4, \dots$
 $\downarrow \downarrow \downarrow \Rightarrow 45 \text{ de nr}$
 $1, 4, \dots$

$45 \cdot 45 = 2025$

3) 243 $243 = 3^5$

$100000 \Rightarrow 2^5 - 1 = 31 \text{ de nr}$

4) $M = \{1, 2, \dots, 10\}$ $\{1, 2, \dots\}$

$1, 2, 3, 4, 5, 6, 7 \rightarrow 8, 9, 10$

$f(A) = M \setminus A$ $f: A \rightarrow B$

5) $9, 9, \dots$
 $\downarrow \downarrow$

$10, 9, 9, 9, 9, 9$
 $9^5 \cdot 10$

6) $9, 10, 10, 1, 1$

$10^3 \cdot 9^2$
 900

A $\equiv \equiv \equiv \equiv \equiv$

B $\equiv \equiv \equiv \equiv \equiv$ C

7) $5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7$

$5 \cdot 3 \cdot 2 \cdot 2 = 5 \cdot 3 \cdot 2 \cdot 2 = 60$

$6 \cdot 5 \cdot 4 \cdot 5$

$24 \cdot 25 = 600$

8) $A \rightarrow B: 6$
 $B \rightarrow C: 5$

$25 \cdot 24 = 600$
 60